



⇒ "Electricity is the silent force that drives chemical change."

Electrochemistry :-

It is the branch of chemistry that deals with the interconversion of electrical energy and chemical energy.

↳ The study of chemical reactions involving electricity is called Electrochemistry.

Key Terms

- **Redox Reaction** : Reaction involving both oxidation and reduction.
- **Oxidation** : Loss of electrons.
- **Reduction** : Gain of electrons.
- **Electrode** : A conductor through which current enters or leaves electrolyte.
- **Electrolyte** : A substance which conducts electricity in molten or aqueous state.

Electrochemical Cell

A device which converts chemical energy into electrical energy, through spontaneous redox reactions.

Types of Cells :

1. **Galvanic (Voltaic) Cell** - Generates electricity.
2. **Electrolytic Cell** - Consumes electricity.

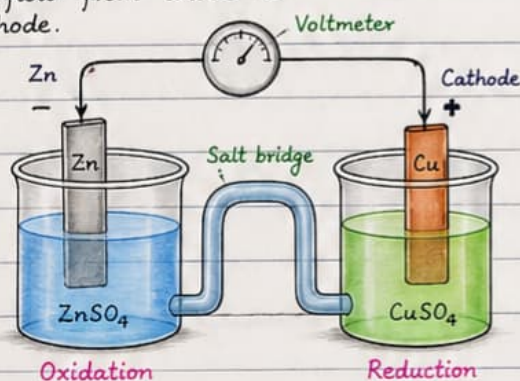
Remember

Galvanic → Chemical to Electrical

Electrolytic → Electrical to Chemical

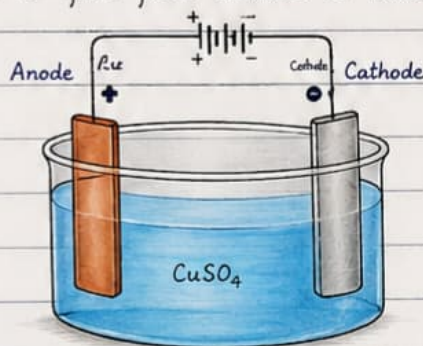
1. Galvanic Cell (Voltaic Cell)

- Converts chemical energy into electrical energy.
- Spontaneous redox reaction occurs.
- e^- flow from anode to cathode.



2. Electrolytic Cell

- Converts electrical energy into chemical energy.
- Non-spontaneous redox reaction occurs.
- e^- flow from cathode to anode.

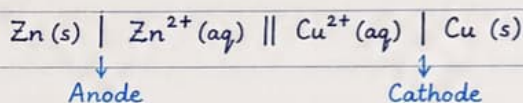


Cell Notation (Galvanic Cell)

General form :

Anode | Anode solution || Cathode solution | Cathode

Example :



Standard Conditions

- ✓ Concentration of ions = 1 M
- ✓ Pressure of gases = 1 atm
- ✓ Temperature = 298 K (25°C)



Applications

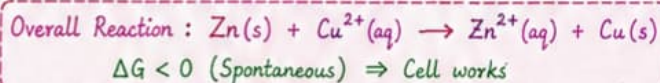
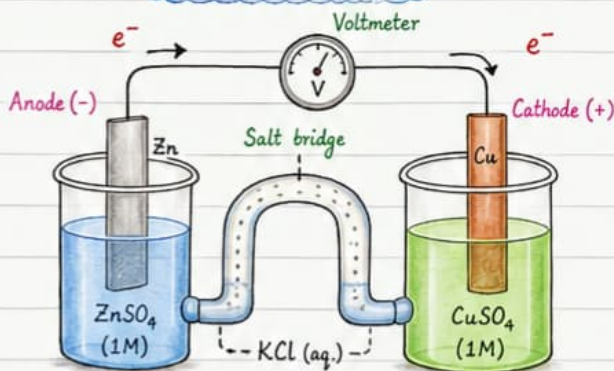
- * Batteries
- * Corrosion prevention
- * Electroplating
- * Extraction of metals



Galvanic Cell (Voltaic Cell)

- Converts chemical energy into electrical energy.
- Based on spontaneous redox reaction.
- Anode (-) : Oxidation occurs.**
- Cathode (+) : Reduction occurs.**
- Electrons flow from anode to cathode through external circuit.
- Salt bridge completes the circuit by allowing ion flow.

Daniell Cell (Example)

Cell Potential (E_{cell})

The potential difference between two electrodes of a cell.

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log Q \quad \text{at } 298 \text{ K}$$

E_{cell}° = Standard cell potential

n = Number of electrons transferred

Q = Reaction quotient

Standard Cell Potential (E_{cell}°)

- Measured under standard conditions.
- All aqueous ions = 1 M
- Pressure of gases = 1 atm
- Temperature = 298 K (25°C)
- Expressed in volt (V).

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

Note :

Higher E° value $\rightarrow$ Stronger tendency for reduction.

Nernst Equation

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log Q \quad \text{at } 298 \text{ K}$$

Where,

E_{cell} = Cell potential under non-standard conditions

E_{cell}° = Standard cell potential

n = No. of electrons transferred

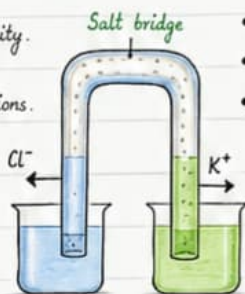
Q = Reaction quotient

Salt Bridge

- U-tube containing inert electrolyte (KCl / KNO_3).
- Maintains electrical neutrality.
- Allows migration of ions.
- Prevents mixing of two solutions.

Functions :

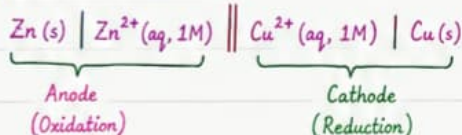
- Completes the circuit.
- Maintains charge balance.
- Keeps solutions electrically neutral.



Cell Representation

- Represented by half-cells using the following convention.
- Anode written on the left, cathode on the right.
- Single vertical line (|) $\rightarrow$ Phase boundary
- Double vertical line (||) $\rightarrow$ Salt bridge

Example : Daniell Cell



Rules

- Anode on left
- Cathode on right
- Inert electrode used for gases.
- Concentration mentioned.

Standard Electrode (Reduction Potential) Series (At 298 K)

Electrodes arranged in increasing order of E° value.

Strong Reducing Agent (-ve E°)	K^+/K	Ca^{2+}/Ca	Na^+/Na	Mg^{2+}/Mg	Al^{3+}/Al	Zn^{2+}/Zn	Fe^{2+}/Fe	Cu^{2+}/Cu	Ag^+/Ag	Au^{3+}/Au	Strong Oxidising Agent (+ve E°)
	-2.93	-2.87	-2.71	-2.37	-1.66	-0.76	-0.44	+0.34	+0.80	+1.50	

- Metals on the left are more reactive.
- They act as reducing agents.

Higher E°
More tendency to gain electrons
(Stronger oxidising agent)

- Metals on the right are less reactive.
- They act as oxidising agents.



Quick Tips

- Flow of electrons : Anode $\rightarrow$ Cathode
- Flow of current : Cathode $\rightarrow$ Anode
- Oxidation : Loss of e^-
- Reduction : Gain of e^-

Key Takeaway

Electrochemistry connects chemical reactions with electrical energy through redox reactions. It is the foundation of batteries, corrosion protection and many industrial processes.

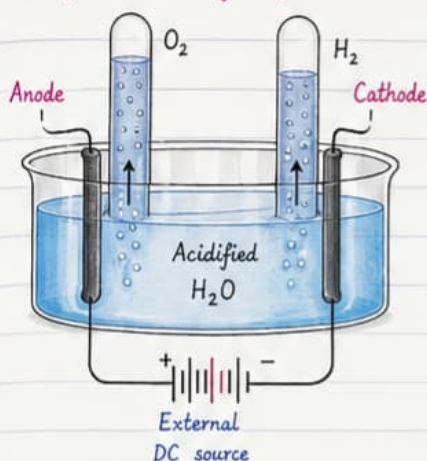




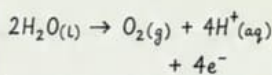
Electrolytic Cell

- Converts electrical energy into chemical energy.
- Non-spontaneous redox reactions occur.
- Requires external source of electrical energy.

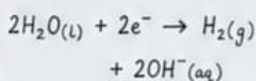
Example : Electrolysis of water



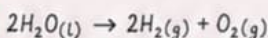
At Anode (+):



At Cathode (-):



Overall reaction:



Faraday's Laws of Electrolysis

- First Law** : The mass of substance deposited or liberated at an electrode is directly proportional to the quantity of electricity passed.

$$m \propto Q \quad \text{or} \quad m = ZQ$$

where,

m = mass of substance (g)

Q = quantity of electricity (C)

Z = electrochemical equivalent (g C^{-1})

- Second Law** : When the same quantity of electricity is passed through different electrolytes, the masses of substances deposited or liberated are proportional to their equivalent weights.

$$\frac{m_1}{m_2} = \frac{E_1}{E_2}$$

where, E = Equivalent weight

Conductance in Electrolytic Solutions

- Electrolytic solutions conduct electricity due to the movement of ions.
- Factors affecting conductance (κ):

- Nature of electrolyte** - Strong electrolytes have higher κ .
- Concentration** - κ decreases with dilution.
- Temperature** - κ increases with rise in temperature.
- Nature of solvent** - Higher dielectric constant $\Rightarrow$ Higher κ .

Types of Conductance

Specific Conductance (κ)

- Conductance of 1 cm^3 of solution with 1 cm distance between electrodes.
- Unit : S cm^{-1}

$$\kappa = \frac{1}{R} \times \frac{l}{A}$$

where, R = Resistance (Ω)

l = distance between electrodes (cm)

A = area of cross-section (cm^2)

Molar Conductance (Λ_m)

- Conductance of all the ions produced from 1 mole of electrolyte in solution.
- Unit : $\text{S cm}^2 \text{ mol}^{-1}$

$$\Lambda_m = \frac{\kappa \times 1000}{C}$$

where, C = Molar concentration (mol L^{-1})

Electrochemical Series

- Arrangement of electrodes in the order of increasing standard reduction potential (E°).
- More negative $E^\circ \rightarrow$ Stronger reducing agent.
- More positive $E^\circ \rightarrow$ Stronger oxidising agent.

Some important electrodes (Standard at 298 K)

Electrode	E° (V)	
Li^+/Li	-3.04	↑ More negative E° Stronger reducing agent
Zn^{2+}/Zn	-0.76	
Fe^{2+}/Fe	-0.44	
H^+/H_2	0.00	
Cu^{2+}/Cu	+0.34	↓ More positive E° Stronger oxidising agent
Ag^+/Ag	+0.80	
Au^{3+}/Au	+1.50	

Applications of Electrochemistry

- Batteries and Fuel Cells** - Provide portable electrical energy.
- Corrosion Prevention** - Galvanisation, cathodic protection.
- Electroplating** - Coating of metals for decoration and protection.
- Extraction of Metals** - e.g., electrolytic extraction of Al, Na, Mg.
- Electrochemical Sensors** - pH meter, glucose sensor, gas sensors.



Quick Revision ★

- ✓ Electrochemistry = Relationship between electricity and chemical change.
- ✓ Galvanic cell : Chemical $\rightarrow$ Electrical | Electrolytic cell : Electrical $\rightarrow$ Chemical
- ✓ $E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n} \log Q$ (at 298 K)
- ✓ $m = ZQ$ (Faraday's 1st law) | $\frac{m_1}{m_2} = \frac{E_1}{E_2}$ (Faraday's 2nd law)
- ✓ Understand cell diagrams, Nernst equation, conductance & applications well.

Concept Booster : Always write anode on left and cathode on right in cell representation.

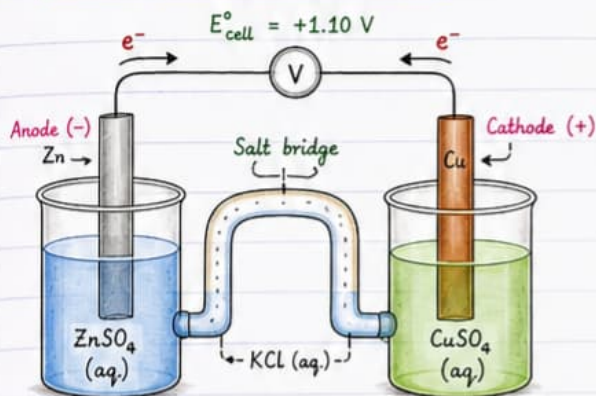




Electrochemical Cells

- Devices in which chemical energy is converted into electrical energy.

Daniell Cell (Zn-Cu Cell)



Anode (Oxidation) : $\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$

Cathode (Reduction) : $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)}$

Overall Reaction : $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

$$= (+0.34) - (-0.76) = +1.10 \text{ V}$$

Electrode Potential (E°)

- Tendency of an electrode to gain or lose electrons.
- Measured with respect to SHE.
- More +ve $E^\circ \rightarrow$ stronger oxidising agent.
- More -ve $E^\circ \rightarrow$ stronger reducing agent.

Standard Electrode Potentials at 298 K

Half Cell Reaction	E° (V)	
$\text{Li}^+(\text{aq}) + \text{e}^- \rightarrow \text{Li(s)}$	-3.04	↑ More positive E° Stronger oxidising agent
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn(s)}$	-0.76	
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe(s)}$	-0.44	
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$	0.00	
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)}$	+0.34	
$\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag(s)}$	+0.80	↓ More negative E° Stronger reducing agent
$\text{Au}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Au(s)}$	+1.50	

Frequently Confused Points

- ✓ In galvanic cell : Anode (-), Cathode (+)
In electrolytic cell : Anode (+), Cathode (-)
- ✓ Electrons flow from anode $\rightarrow$ cathode in external circuit.
- ✓ Salt bridge maintains electrical neutrality.
- ✓ SHE has $E^\circ = 0.00 \text{ V}$ (reference electrode).

Galvanic (Voltaic) Cell

- Spontaneous redox reaction.
- $\Delta G < 0$, $E^\circ_{\text{cell}} > 0$
- Anode (-), Cathode (+)

Electrolytic Cell

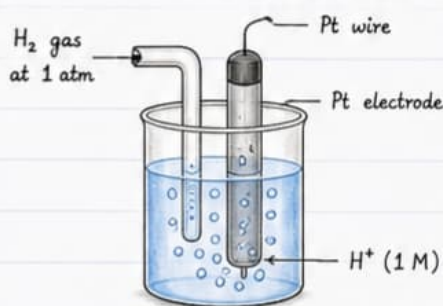
- Non-spontaneous redox reaction.
- $\Delta G > 0$, $E^\circ_{\text{cell}} < 0$
- Anode (+), Cathode (-)

Standard Electrode (Half Cell)

A half cell in which the electrode is in contact with its 1 M (1 mol L^{-1}) solution at 298 K and 1 atm pressure.

Ex. $\text{Zn} | \text{Zn}^{2+} (1 \text{ M})$ (Standard hydrogen electrode is the reference electrode)

Standard Hydrogen Electrode (SHE)



$$E^\circ_{\text{SHE}} = 0.00 \text{ V}$$

(By convention)

Nernst Equation (At 298 K)

For the cell reaction,



$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n} \log Q$$

where,

E_{cell} = Cell potential under non-standard conditions

E°_{cell} = Standard cell potential

n = Number of electrons transferred

Q = Reaction quotient

At 298 K,

$$0.0591 = \frac{2.303 RT}{F}$$

Significance of Electrochemistry

- ◆ Provides a method to generate electricity (batteries, fuel cells).
- ◆ Prevents corrosion (galvanisation, sacrificial protection).
- ◆ Extraction of metals (electrolytic refining).
- ◆ Electroplating for protection and decoration.
- ◆ Electrochemical sensors used in medical & environmental analysis.

Quick Summary

- ★ Electrochemistry = Study of conversion between chemical and electrical energy.
- ★ Galvanic cell $\rightarrow$ Generates electricity ($\Delta G < 0$).
- ★ Electrolytic cell $\rightarrow$ Consumes electricity ($\Delta G > 0$).
- ★ $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$
- ★ Nernst equation relates E_{cell} with concentration.



Key Takeaway ★ Electrochemistry is the bridge between chemistry and electricity - Powering devices, protecting materials and enabling modern technology.

